Manmeet Kaur Sodhi

Soft Interfaces | Particle Characterization | Scientific Instrumentation

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Scientific Profile

PhD researcher in soft matter and polymer colloids at TU Darmstadt and the Institut Laue-Langevin, combining single-particle nanomechanics, interfacial science and neutron/X-ray scattering to study soft particles across length scales. My work connects microgel composition, charge and deformability to adsorption, confined-film structure and macroscopic foam response. I am interested in R&D, application science and scientific instrumentation roles involving colloids, dispersions, interfaces, particle characterization and formulation-relevant soft materials.

Selected Research Contributions

- Established a multiscale characterization framework linking microgel composition, particle mechanics, interfacial organization and foam-film structure.
- Studied six matched PNIPAM microgel formulations to resolve the effects of crosslinking density and initiator-derived charge.
- Developed AFM force-spectroscopy and nanomechanical mapping workflows for highly soft polymer particles in liquid.
- Combined DLS, zeta potential, AFM, SANS/SAXS, neutron reflectometry and optical image analysis to connect particle-, film- and foam-scale structure.
- Led four large-scale facility proposals as main proposer and contributed to collaborative neutron/synchrotron experiments.

Methods & Technical Expertise

Soft-matter systems	Polymer colloids, microgels, aqueous dispersions, soft interfaces, thin films, foams, structure–property relationships.
Particle characterization	AFM force spectroscopy, quantitative imaging, DLS, zeta potential, microscopy, optical image analysis.
Scattering & reflectometry	SANS, SAXS, neutron reflectometry, XRR, XRD, scattering-data reduction and model interpretation.
Surface / materials methods	Raman spectroscopy, SEM, ellipsometry, interferometry, thin-film and interfacial characterization.
Data & software	Python, Origin, Gwyddion, QtiPlot, GRASP, COMSOL, LaTeX, Inkscape, MS Office.

Research Experience

Technical University of Darmstadt & Institut Laue-Langevin *Nov 2023 – Present*
Doctoral Researcher, Soft Matter at Interfaces Darmstadt / Grenoble

- Investigate how PNIPAM microgel composition, crosslinking density and initiator-derived charge govern particle mechanics, interfacial organization and foam stability.
- Designed and characterized six matched microgel formulations using DLS, zeta potential, AFM force spectroscopy, SANS/SAXS, neutron reflectometry and optical image analysis.
- Developed AFM force-spectroscopy and nanomechanical mapping workflows for soft polymer particles in liquid.
- Planned and analysed large-scale neutron-scattering experiments at ILL, including sample preparation, measurement strategy, data reduction and model interpretation.
- Built Python workflows for force-curve analysis, nonlinear fitting, statistical comparison, scattering-data visualization and optical image analysis.
- Preparing first-author manuscript on microgel-containing foams, linking particle mechanics to confined-film structure.

European Synchrotron Radiation Facility & CEA *Feb 2023 – Aug 2023*
Master's Research Thesis Grenoble

- Investigated graphene and hexagonal boron nitride growth on liquid and single-crystal copper using operando imaging and synchrotron X-ray methods.
- Analysed XRD and XRR data to resolve thin-film growth, interfacial structure and material organization.
- Developed Python-based video-analysis workflows to quantify graphene-flake motion, rotation and spatial organization.
- Worked with beamline scientists on experimental setup, data interpretation and comparison with physical models of Marangoni-driven assembly.

CNRS – Institut Néel *Feb 2022 – Jun 2022*
Graduate Research Intern Grenoble

- Improved reproducibility of nanowire-device fabrication through process optimization.
- Characterized nanostructures using SEM, ellipsometry and interferometry.

Large-Scale Facility & Advanced Characterization Experience

Institut Laue-Langevin, Grenoble

- Led large-scale neutron facility proposals as main proposer and contributed to collaborative experiments on soft matter, polymer colloids and interfacial systems.
- Performed SANS and neutron-reflectometry work involving microgel-containing foams, foam films and confined soft-matter structures.
- Worked on sample preparation, measurement planning, beamtime coordination, data reduction, model interpretation and scientific reporting.

European Synchrotron Radiation Facility, Grenoble

- Performed synchrotron X-ray reflectivity and grazing-incidence X-ray diffraction measurements for interfacial and thin-film systems.
- Analysed structural evolution using X-ray data and complementary microscopy/spectroscopy methods.

Selected Publications

- **Sodhi, M. K.**; Robertson, H.; Schneider, C.; Chiappisi, L.; Soltwedel, O.; von Klitzing, R. “Multiscale Structure of Microgel-Containing Foams: From Particle Mechanics to Confined Films.” *Manuscript in preparation*.
- Prabhu, M. K.; Rein, V.; Ghanadan, N.; **Sodhi, M. K.**; et al. “Marangoni-Driven Active Assembly of Graphene Flakes on Liquid Metal Surfaces.” *Submitted*.
- Ghanadan, N.; Rein, V.; Gao, H.; et al.; **Sodhi, M. K.**; et al. “Hexagonal Boron Nitride on Liquid and Single-Crystal Copper: Operando X-Ray and Atomistic Insights into Growth and Interfacial Structure.” *Advanced Science* (2026).
- Poudel, B.; Ritzert, P.; Robertson, H.; et al.; **Sodhi, M. K.**; et al. “Comparing Simulated and Synthesized Polymer Brush Profiles.” *Journal of Chemical Physics* **163**, 174906 (2025).
- Buati, N.; Auzelle, T.; John, P.; et al.; **Sodhi, M.**; et al. “AlN Nanowire-Based Vertically Integrated Piezoelectric Nanogenerators.” *ACS Applied Nano Materials* **7** (2024).

Scientific Communication

- Invited Session Chair and Oral Presentation, European Student Colloid Conference, Donostia–San Sebastián, 2026.
- Oral Presentation, UK Neutron and Muon Science and User Meeting, University of Warwick, 2026.
- Oral Presentation, European Colloid and Interface Society Conference, Bristol, 2025.
- Oral Presentation, SoftComp Annual Meeting, Venice-Mestre, 2025.
- Oral Presentation, European Summer School on Scattering Methods Applied to Soft Condensed Matter, Carcans-Maubuisson, 2024.
- Poster presentations at International Liquid Matter 2024, DPG Spring Meeting 2024, Microgels 2024 and ILL Summer School Workshop 2023.

Teaching & Communication Experience

Blue Star Public School — *Jul 2018 – Mar 2020*
Science & Mathematics Teacher, Grades 8–12 — Mandi Gobindgarh, India

- Taught science and mathematics to students in Grades 8–12, preparing structured lessons, assessments and clear explanations adapted to different learning levels.

Private Tutoring — *2019 – 2021*
Physics Tutor, Senior Secondary — Mandi Gobindgarh, India

- Provided one-to-one physics tutoring with progress tracking and individualized explanations adapted to students’ learning needs.

Education

Technical University of Darmstadt — *Nov 2023 – Present*
PhD Researcher, Soft Matter at Interfaces

Université Grenoble Alpes — *2022 – 2023*
Master 2, Nanophysics

Université Grenoble Alpes — *2021 – 2022*
Master 1, Nanophysics and Quantum Physics, Research Track

Guru Nanak Dev University — *2015 – 2018*
Bachelor of Science, Physics

Languages

Punjabi: native | Hindi: fluent | English: C1, IELTS 8.0/9 | French: B1 | German: A1